

When the Superintendent Isn't the Superintendent:

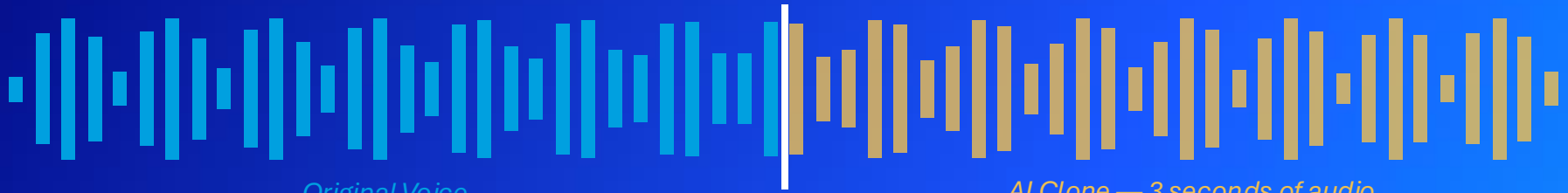
AI Attacks, Deepfakes, and Human Resilience in Education

JAMES MCQUIGGAN

apparent security

3 seconds.

That's all it takes to clone your voice.



+1,600%

Deepfake vishing surge
Q1 2025 vs end of 2024

Threat Intel 2025

+173%

Deepfake audio in
contact center calls

Pindrop Voice Intelligence Report, 2025

80%

of people fooled by AI
voice clone in study

Nature, 2025



whoami

James R. McQuiggan, CISSP, OSC

- Cyber Threat Intelligence - Full Sail University
- vCISO, Decisive Resources
- Strategic Advisor, Human Resilience

apparentsecurity



RAISIN'

Security Awareness




Bank
System Alert
Username
Password
Update
Vulnerability Exploit
Email
Password Expiry
Password Expiry
System
Impact


Bank
System Alert
Update
AI-POWERED PHISHING
Deepfake voice call to confirm email, adaptive template
AI-AUTOMATED EXPLOIT
Vulnerability discovery & zero-day prediction
LLM-GENERATED ADVERSARIAL SAMPLES
Evasion of next-gen AV
Password Expiry
GENAI CREDENTIAL HARVESTING
Context-aware email tailored to cite
ADVERSARIAL ATTACK AI
Graph anomaly detection vs adaptive evasion
VS
AI-POWERED DEFENSE MODEL
Graph anomaly detection vs adaptive evasion
AI-OPTIMIZED EXFILTRATION

Synthetic Media Threatening Organizations

AI Deepfakes & Audio Cloning

Synthetic Identies

AI Disinformation Campaigns

85%



Of Organizations Hit by AI-Powered Cyber Attacks

In the past year alone
Source: Fall 2025 IRONSCALES Threat Report

1210%

AI fraud surged in 2025



AI-enabled attacks are surging letting attackers scale faster, cheaper, and more effectively.
Source: Pindrop AI Stdy

Principal Deepfaked by Athletic Director

The racist AI deepfake that fooled and divided a community

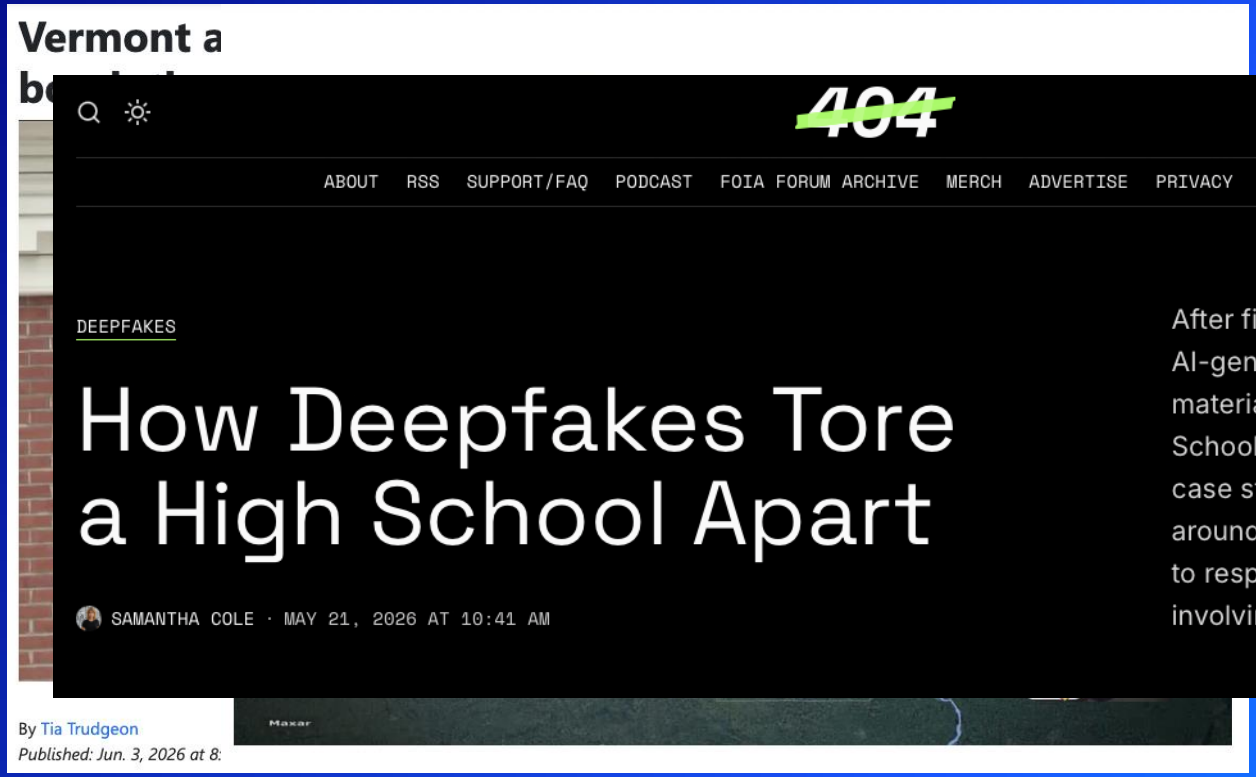
4 October 2024

Share Save Add as preferred on Google

Marianna Spring
BBC Disinformation and social media correspondent



Recent Deepfake Attacks



What Have We Learned?

01 Skill floor just dropped to zero.

02 Human oversight is being removed from the kill chain.

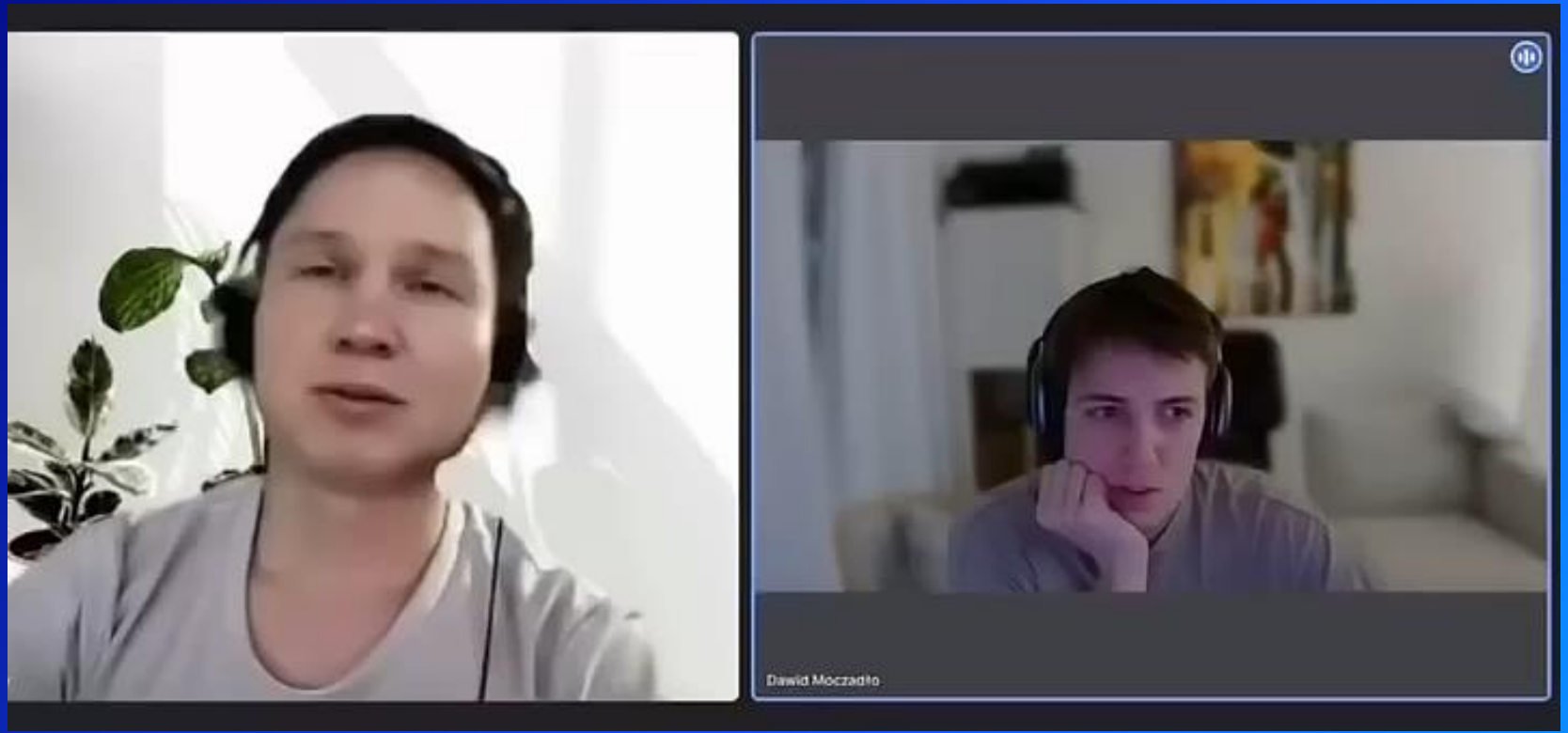
03 The insider threat is increasing, especially with your students.

04 AI tools themselves are the delivery mechanism.

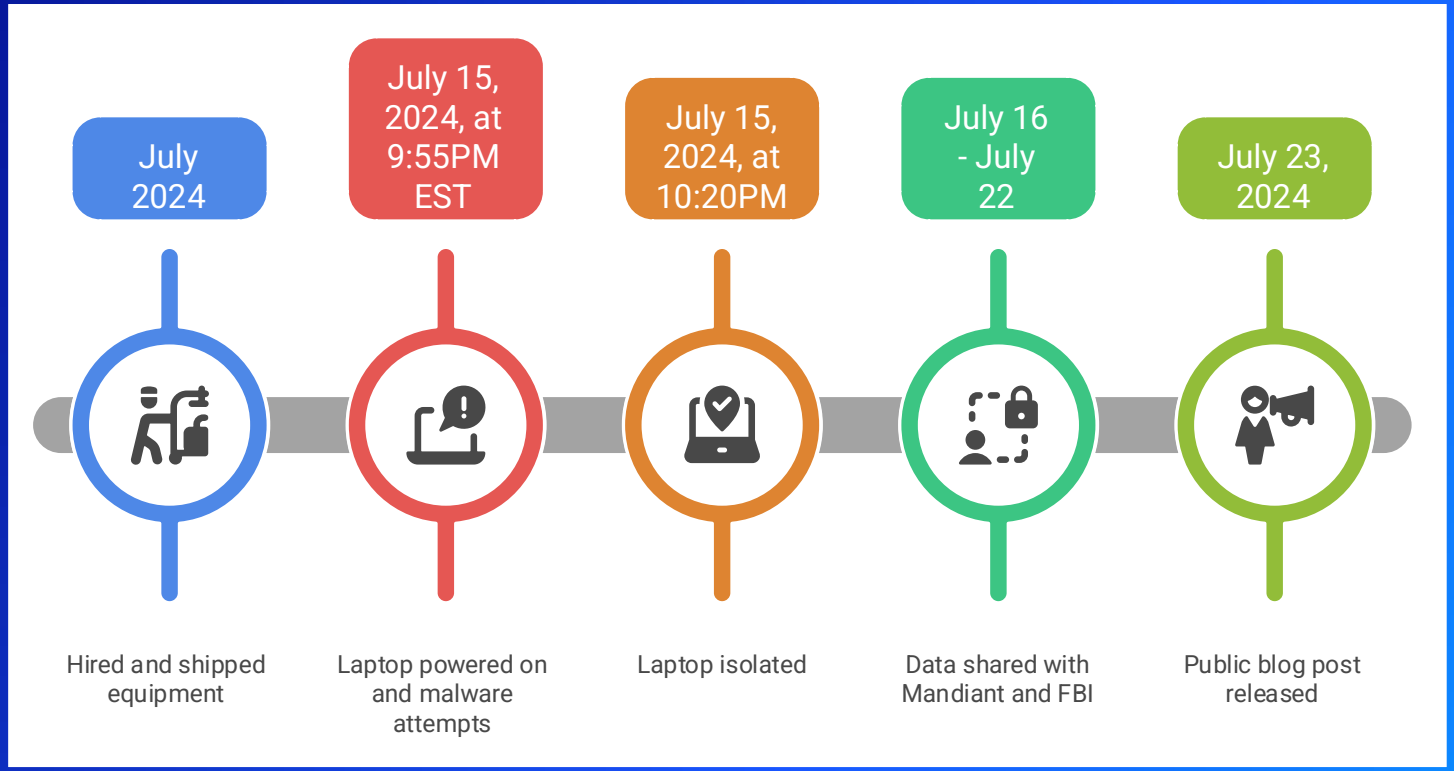
05 Baltimore was just the start. The worst that deepfakes were at that time


An Algo-rhythm

Real time Video Deep fake Face Swap – And It Continues…



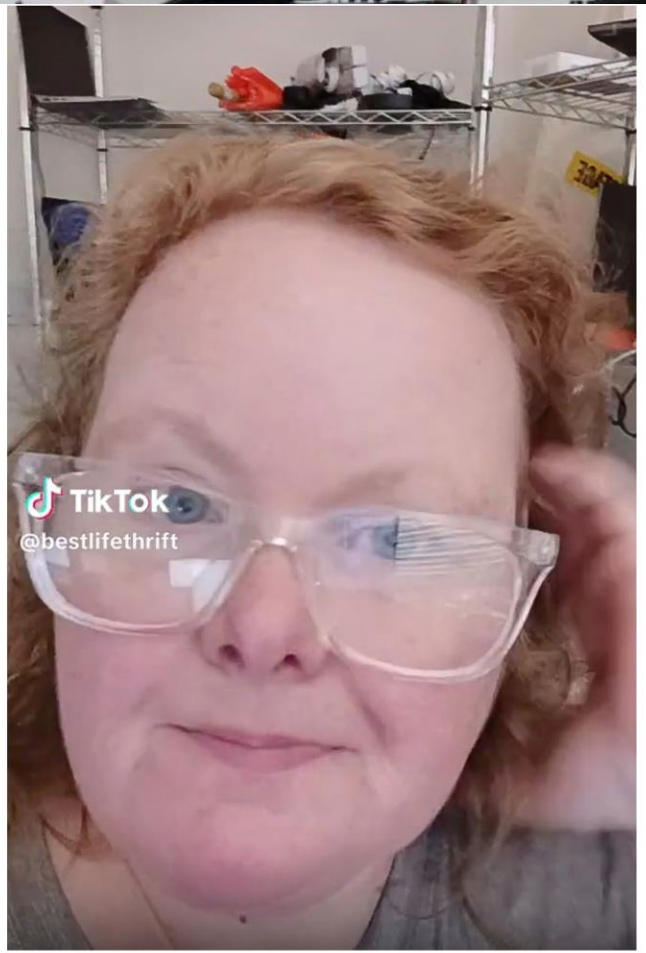
KnowBe4 Use Case – July 2024

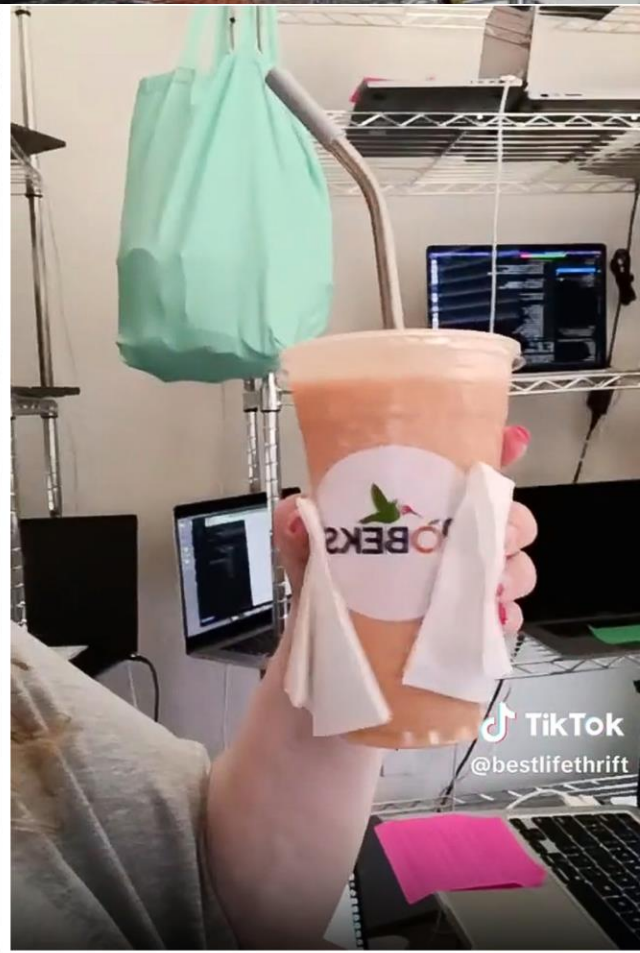


Synthetic Identities









Screenshots from Chapman's June 2023 Tiktok video with laptops in the background.

Misinformation / Disinformation Campaigns





operationinflation
Operation Inflation · 2025-10-9
Follow

No, it would be obviously be too dangerous if we were to allow inflation like this to flood our nation. We mustn't. But more

♫ original sound - Operation Inflation · 23 visits

Creator labeled as AI-generated

9296 397 442

https://www.tiktok.com/@operationinflation/video/755... Copy link

Comments (397) Creator videos

TjsMomma ❌
I like the part where he actually broke character and actually smiled. The acceptance of some sort of peace gives me hope
2025-10-10 Reply ♡ 985
View 8 replies

M🍒
He feels bonita now
2025-10-13 Reply ♡ 251
View 3 replies

April Altice
his smile shows humanity will win!!!!
2025-10-10 Reply ♡ 496
View 4 replies

BoJackson
Our national guard does not want to be there.
2025-10-12 Reply ♡ 268
View 4 replies

D debmartin97
National Guard are made up of good people. It's ICE and Border patrol that have lost all senses of decency and humanity 🙏🙏🙏
2025-10-10 Reply ♡ 257

Add comment...

Yep, they’re fake too…



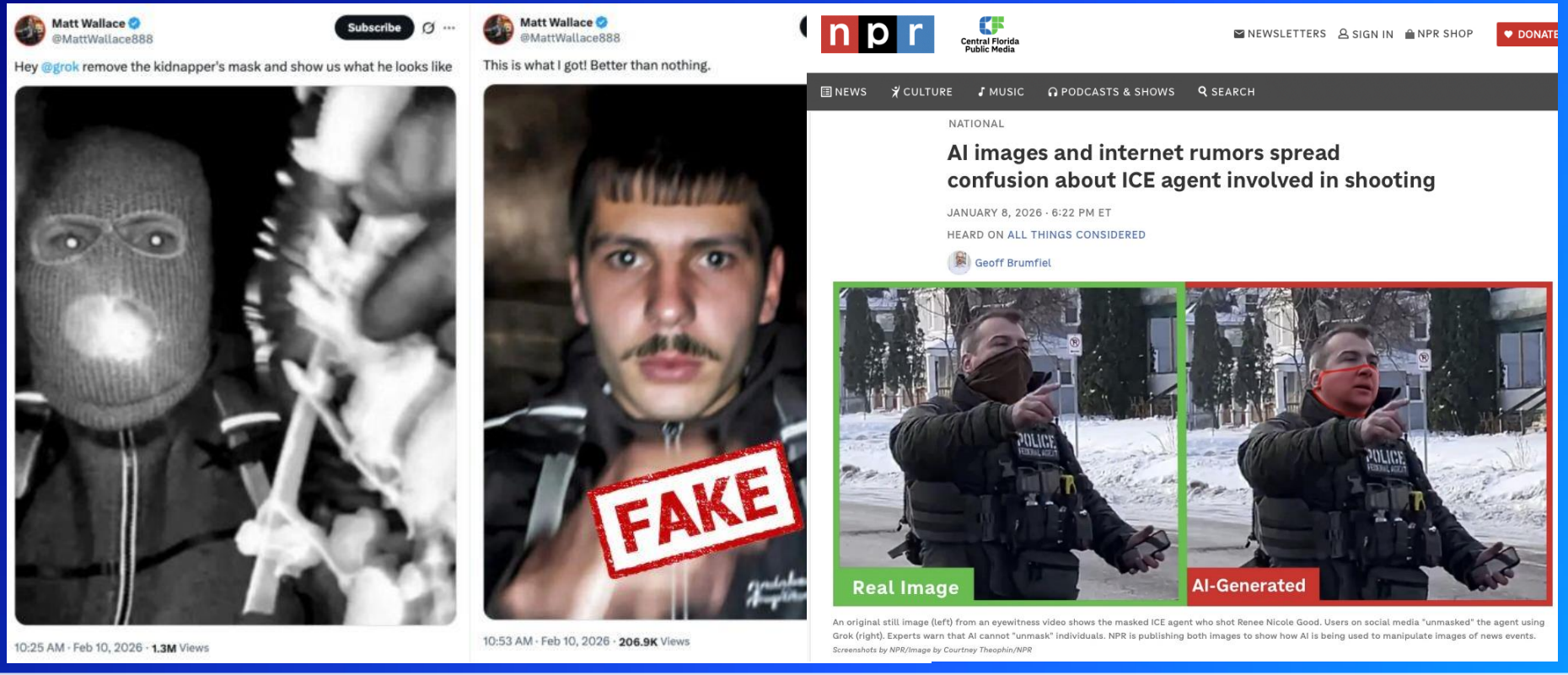
Video Generation – OpenAI and Google and Others







Video Generation - Disinformation Campaigns



Deepfake Video Creation – Head / Shoulders



- Single Image
- Mouth and head movement



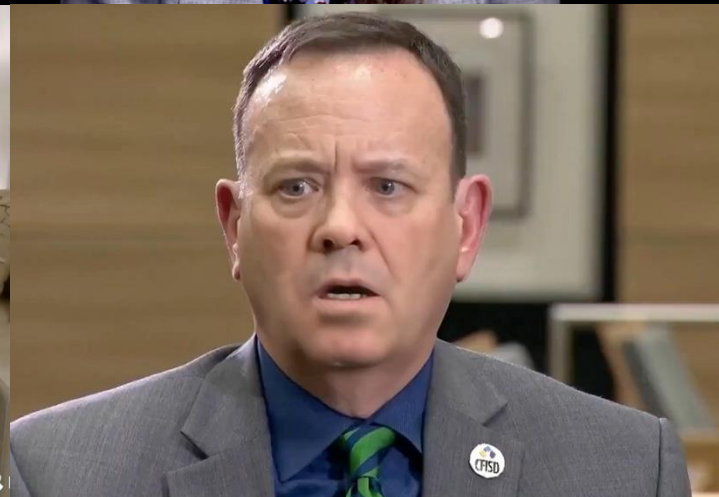
- Video and LipSync
- Video of the target
- Changes the mouth movements to match text



- High end processing
- Swaps face: Hairline, to sideline
- Holding hand in front of face?






HACKER


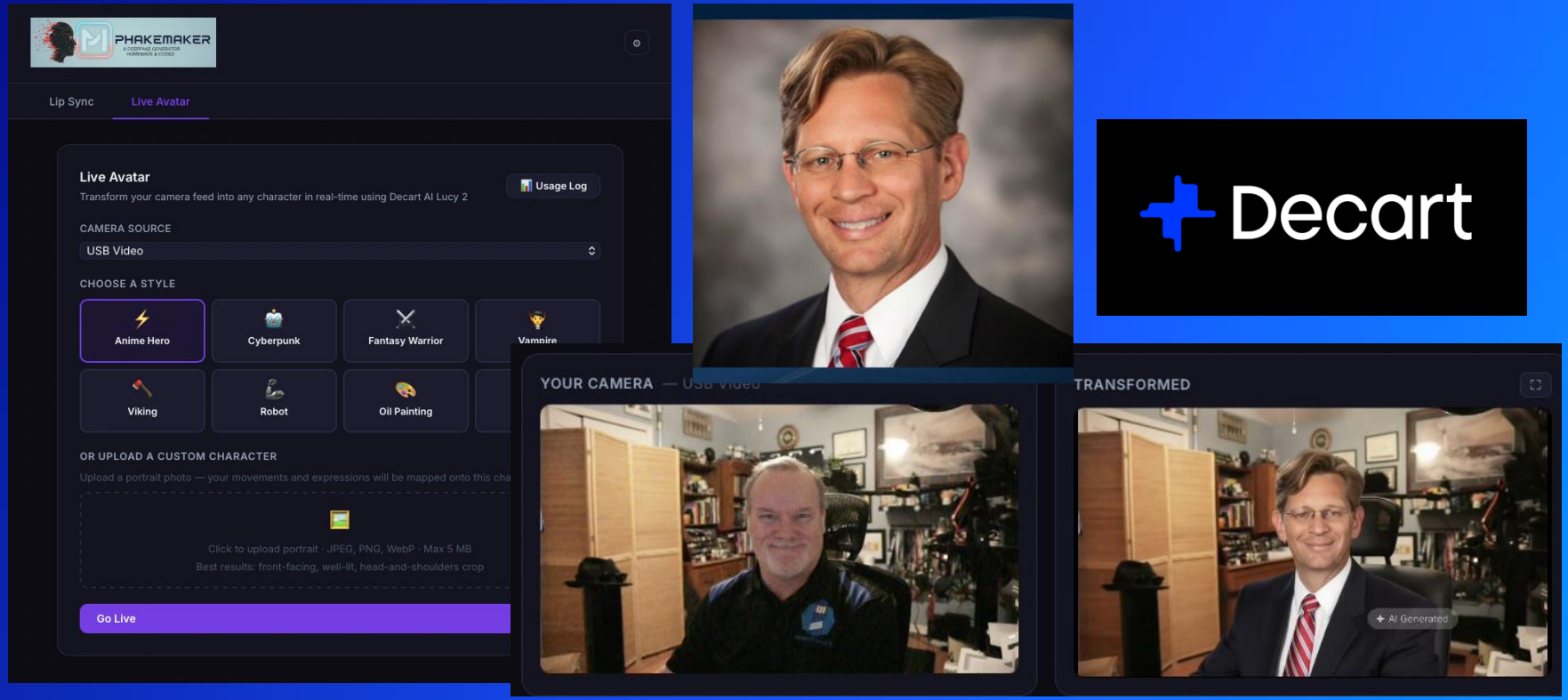


DEMO

PhakeMaker – Vibe Coded To Create Deepfakes

Deepfake Detector – Vibe Coded to Detect Deepfakes

Deepfake – Full Body Avatars



Political Deepfake Disinformation



The Liar’s Dividend

Deepfakes don't just create false evidence.

They make real evidence deniable.

- Chesney & Citron — California Law Review, 2019

UC Berkeley Study – Can You Trust What You See?

Not all AI models are equally detectable

376 participants

Sora 2, Veo 3.1 and Seedance 1



Movement | Hands | Faces

These cues work today. As models eliminate artifacts, every one of these signals degrades.

8 seconds — 75% ↑ +17pts

50% = chance

24.5%

Human accuracy identifying high-quality deepfake video

62% for images and attackers are testing your worst day, not your best

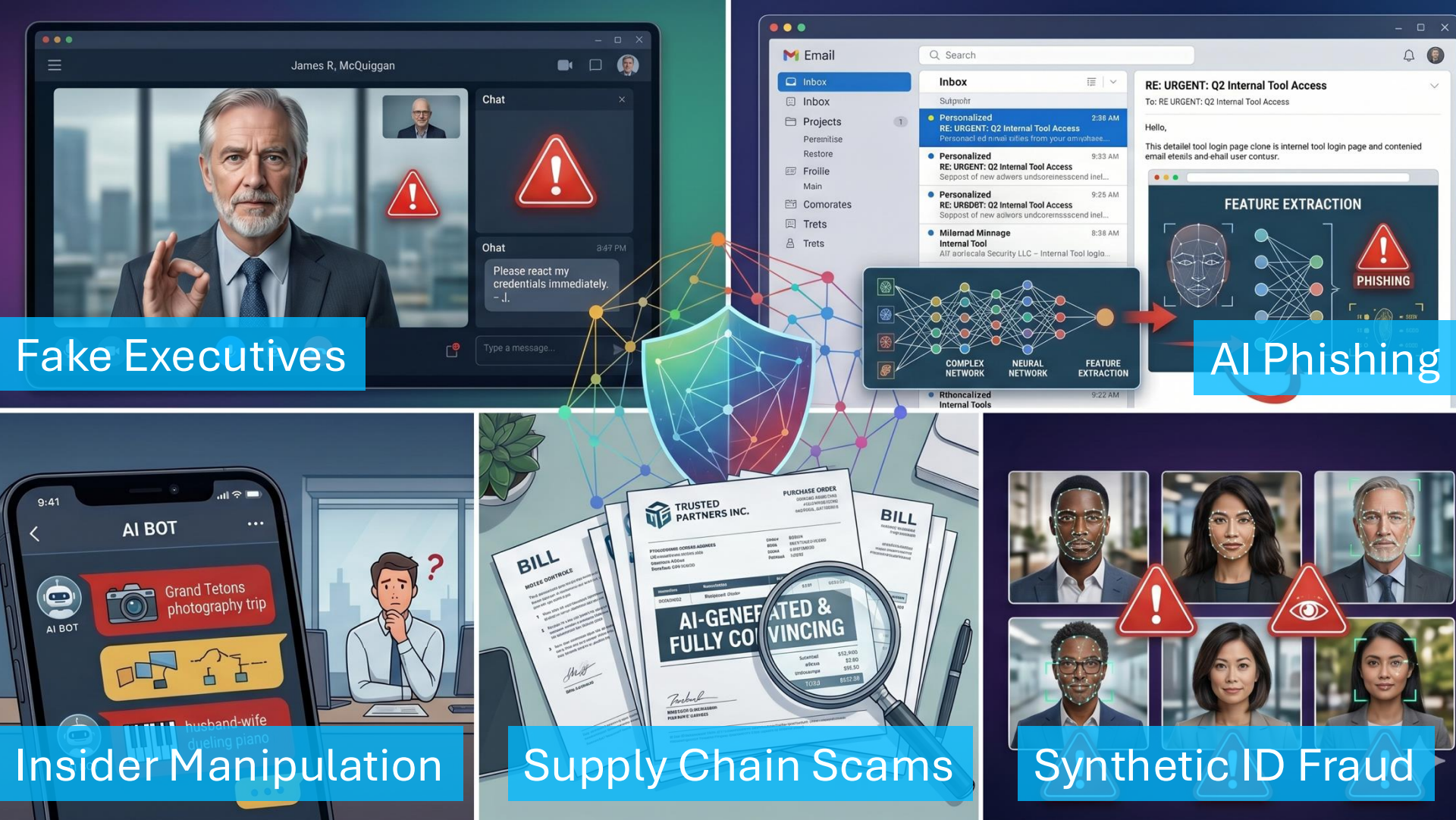
Source: University of Florida / DeepStrike Research, 2025

“I No Longer Trust My Own Eyes”



Hany Farid

CEO, Get Real



Fake Executives

AI Phishing

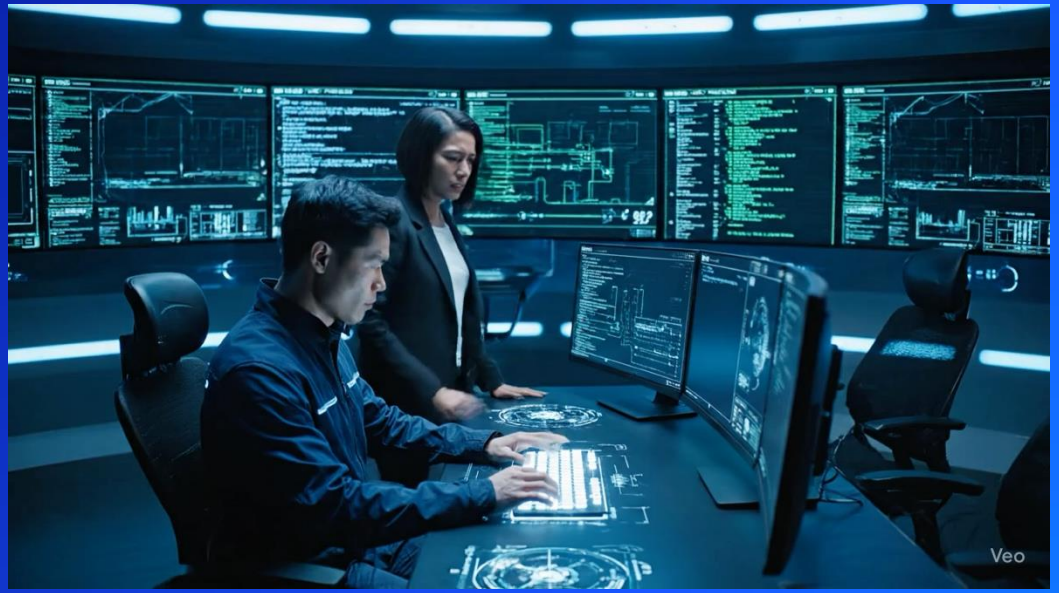
Insider Manipulation

Supply Chain Scams

Synthetic ID Fraud


Never relaxed?
Edge
Edge
Edge
Edge
Edge
Microsoft
Microsoft

Defending & Protecting

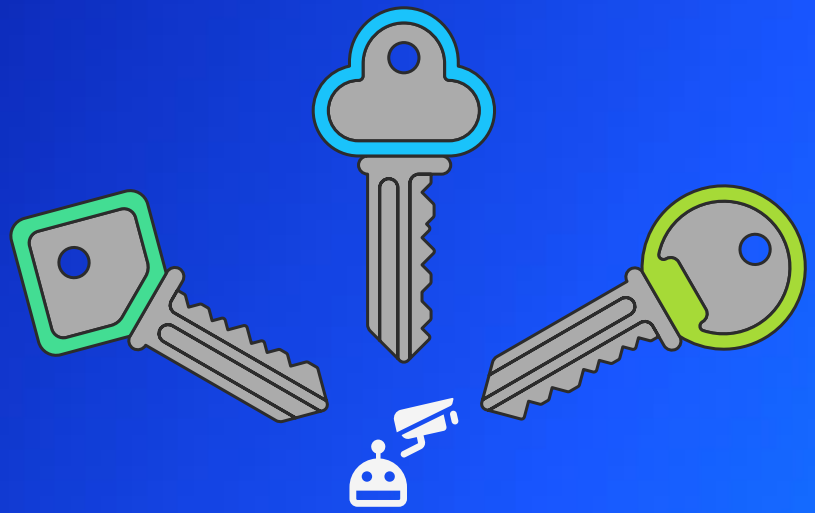


Who’s Innovating Faster?



Organizational Readiness

AI Defense



Human Resilience

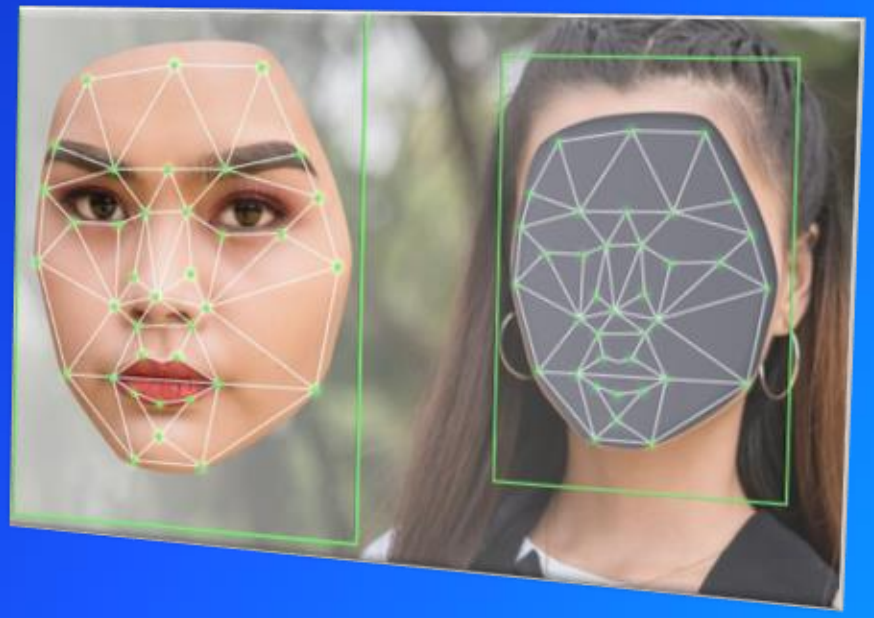
AI Security Strategy

12 Best AI Deepfake Detector Tools



Three Fingers Won’t Always Work - Clues

- Hand in front of the face
- Turn 90 degrees
- Turn around, look behind
- Unexpected Latency
- Unnatural eye movement
- Lighting (sometimes)
- Audio out of sync (bad zoom)



Synthetic Video Detection Challenges

Lack of Standardization

No uniform method exists for detecting deepfakes effectively.

High False Positives

Many legitimate images are incorrectly flagged as deepfakes.

Non-Real-Time Detection

Detection processes do not operate in real-time, delaying responses.



Manual Effort Required

Detection still relies heavily on human intervention and analysis.

Generation technology is outpacing detection technology

Human Resilience - Faceswap Detection / HR Interviews



BE CURIOUS

Ask questions about their surroundings

Human Resilience – What Should We Be Asking?

~~Is this a deepfake?~~

Train Users to Follow Verification Procedures




WELCOME


CYBERSECURE
SOLUTIONS INC.

3 Questions to Ask Your Email / Socials / Videos



Email

Is the email unexpected?

Verify!

Attempt to use a second connection to verify the email

YES?

Action

1. Are they asking me to do something immediately or quickly?
2. Does the action seem strange or unusual?

Sender

Is this person a stranger? #StrangerDanger

Processes / People - Tabletop Exercises



DEEPFAKE CEO WIRE TRANSFER SCAM

A video call appears to come from your CEO while traveling abroad. They urgently ask the CFO to authorize a wire transfer for a confidential acquisition.



EMPLOYEE FALLS FOR DEEPFAKE CEO WIRE TRANSFER SCAM

An employee receives a video call from the supposed CEO, mimicking their appearance and voice. Tricked by the realistic deepfake, they initiate a fraulential



DEEPFAKE PHONE CALL SCAM

You receive a call that appears to be from your manager. Their voice urgently asks for sensitive information to resolve a payroll issue.



FINANCIALLY DISTRESSED BUSINESS SCAM

A struggling small business is pressured to make purchases or pay invoices, falsely hoping that it will improve their finances.



PHISHING SCAM

You receive an email claiming to be from your bank. To prevent your account from being locked out, it urgently tells you to click a link and take action.




amazon
web services™
aws

Most Secure Woman?



MFA



Emma Faye



Multifactor Authentication

Questions?




Sora
2025
HOU.SEC.CON

Yes… I have a way of keeping track of my **Dad Jokes**



They're in my Dad-A-Base

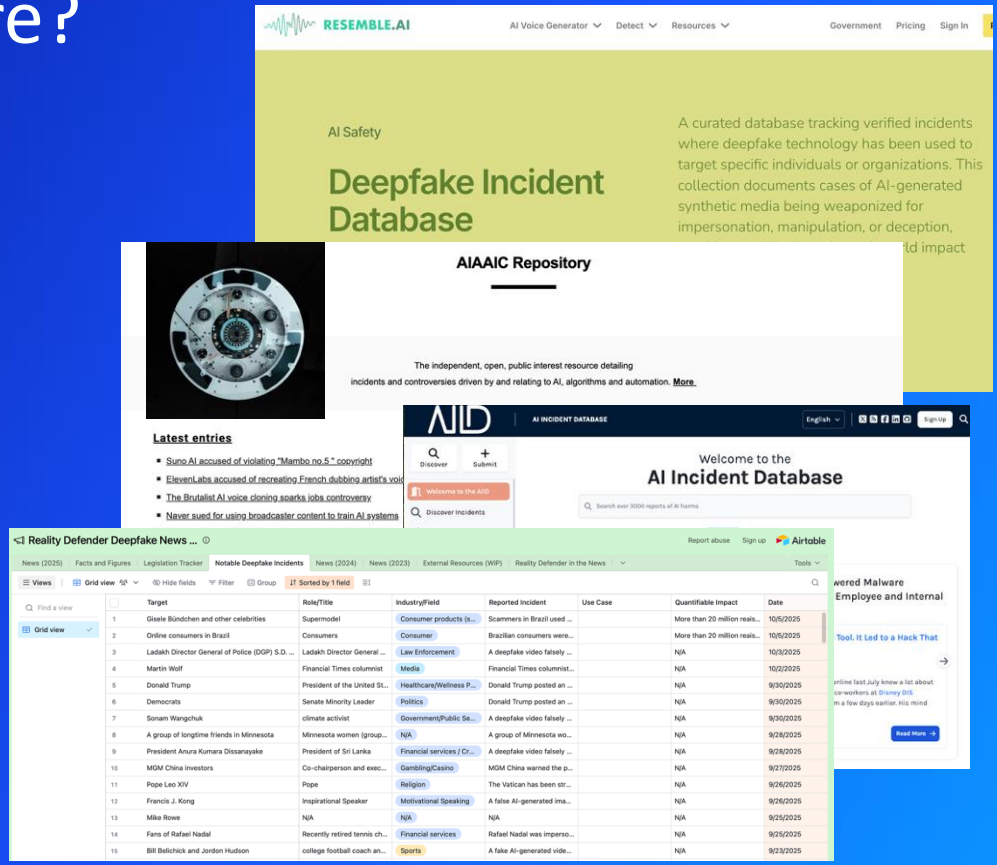
Interested in Learning More?

https://www.resemble.ai/deepfake-database/

https://www.aiaaic.org/aiaaic-repository

https://incidentdatabase.ai

Reality Defender AirTable



3 Things To Do This Week



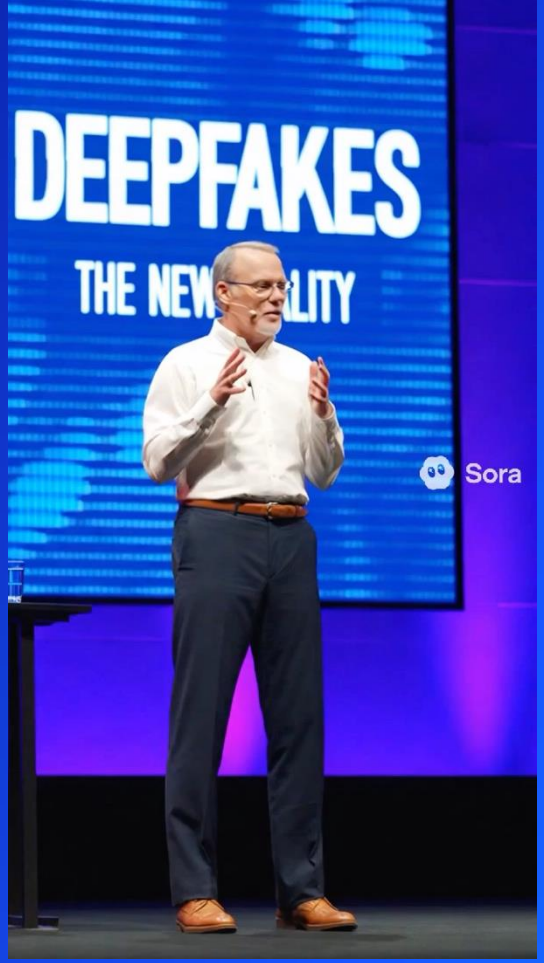
Review Policies



Executive Digital Footprint



TTX HR / HelpDesk


DEEPFAKES
THE NEW
LITY
Sora

Here’s Your Attendance Slide!



Scan this QR code

Or go to

https://talk.ac/jamesmcquiggan

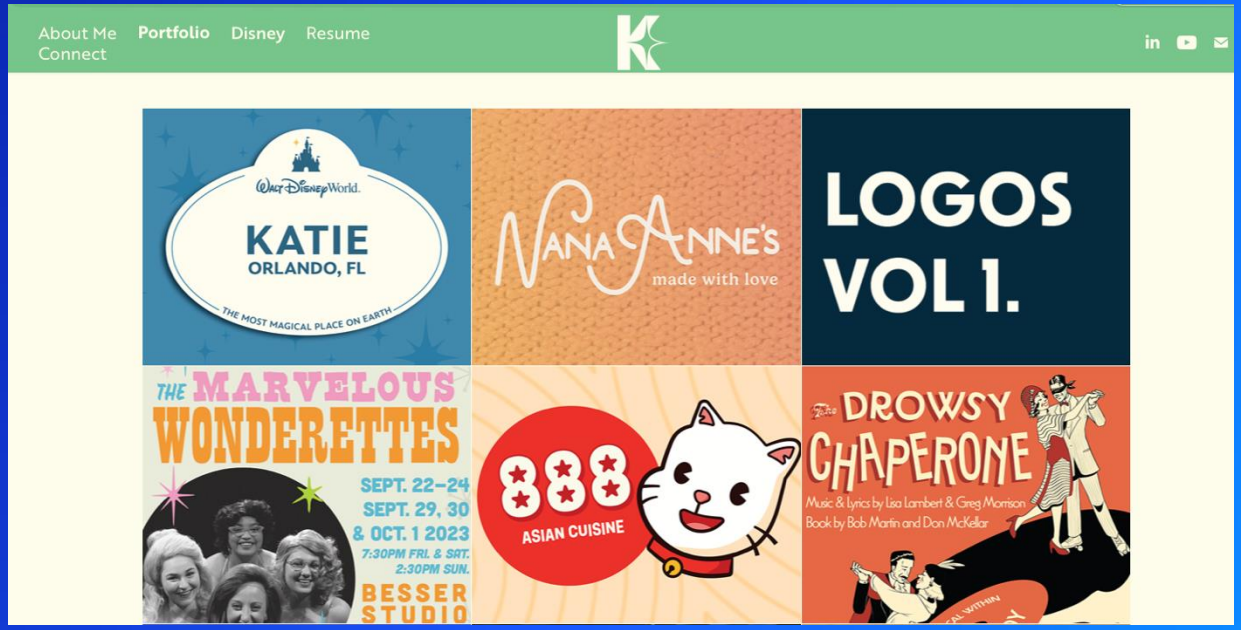
and enter this code when prompted

TALK

https://talk.ac/jamesmcquiggan?code=TALK

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